

Lecture 3:

Accumulation zone processes

Mass balance (MB)

Specific mass balance

Surface facies

Accumulation

Measuring MB

Firn densification

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Accumulation zone processes

Mass balance

Specific mass balance

Surface facies

Accumulation

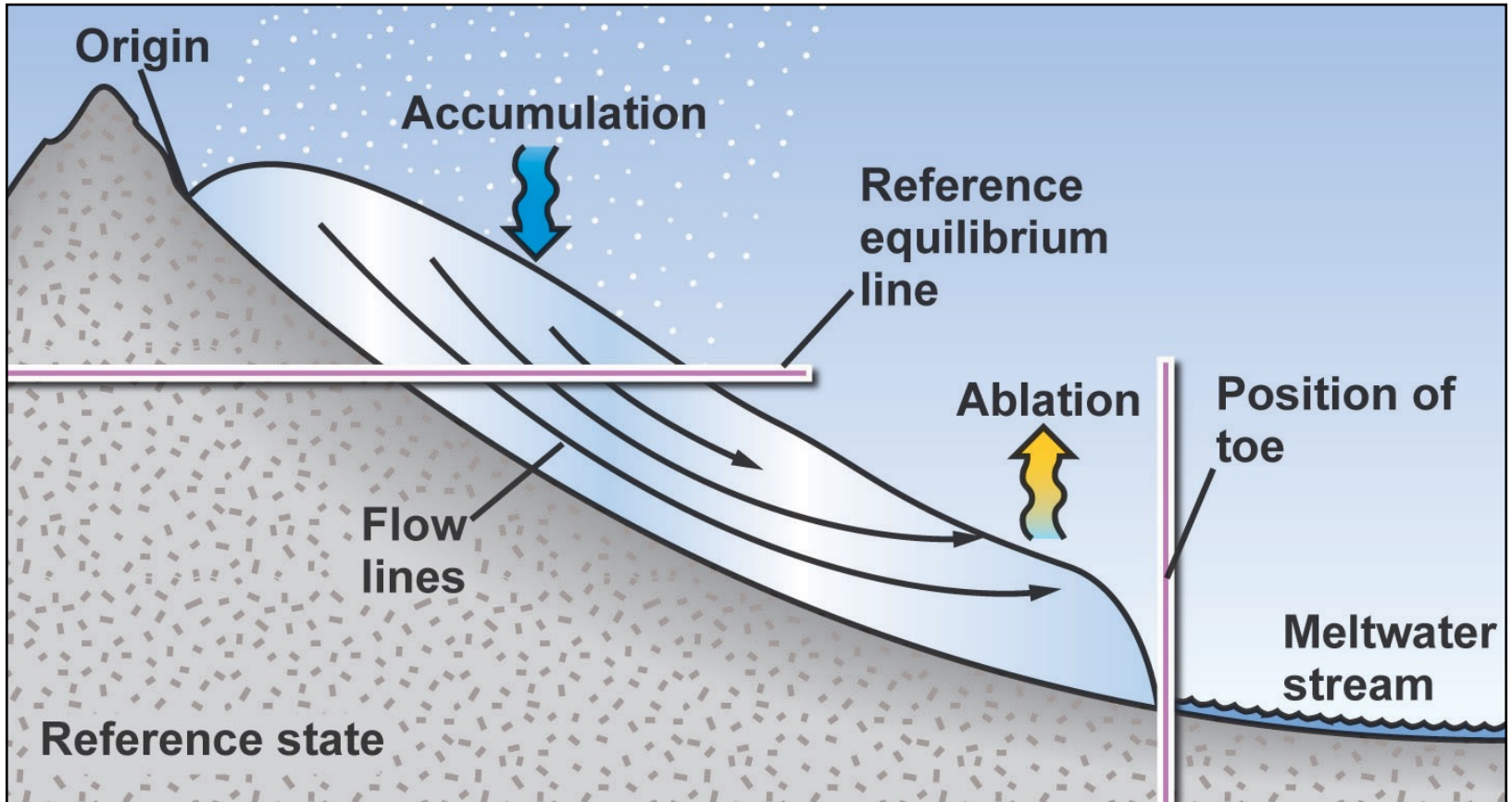
Measuring MB

Firn densification

Discuss office hours.

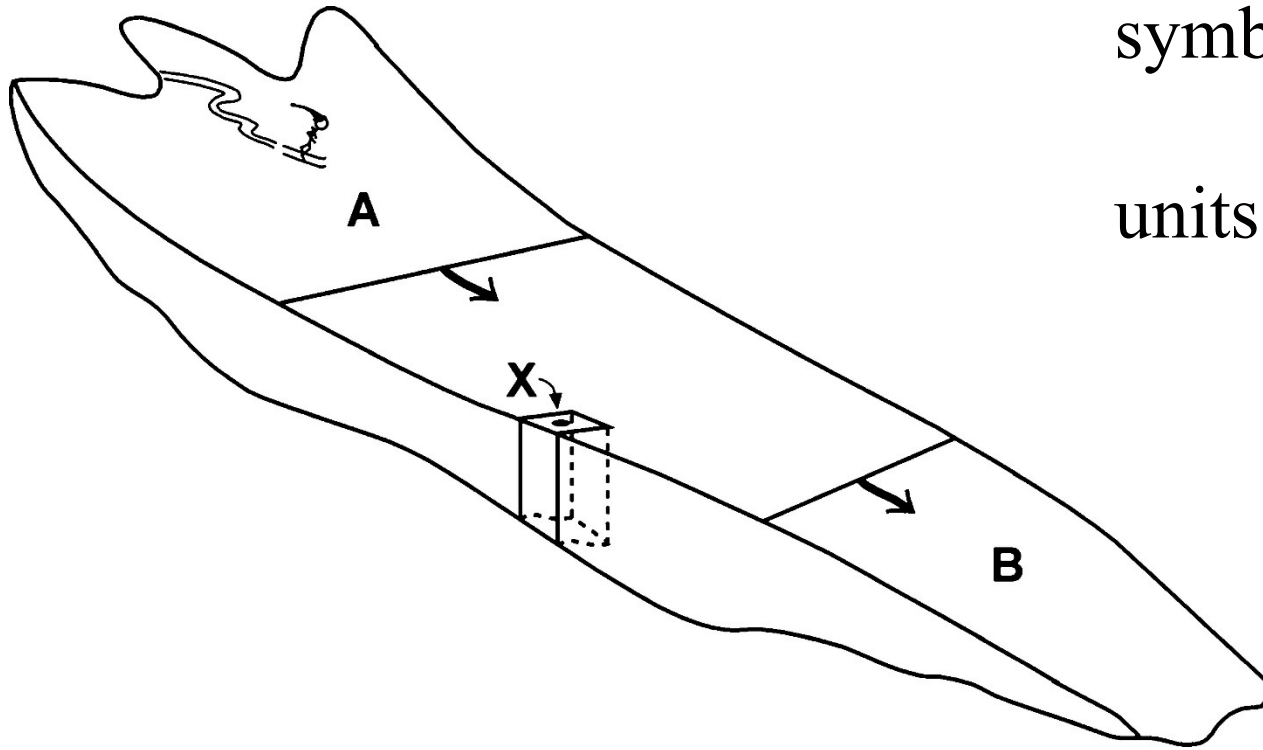
Mass balance

$$\text{Mass balance} = \text{total mass gain} - \text{total mass loss}$$



Specific mass balance

Change in mass per unit area



symbol: b

units: kg m^{-2}

Cuffey and Paterson (2010)

Specific mass balance

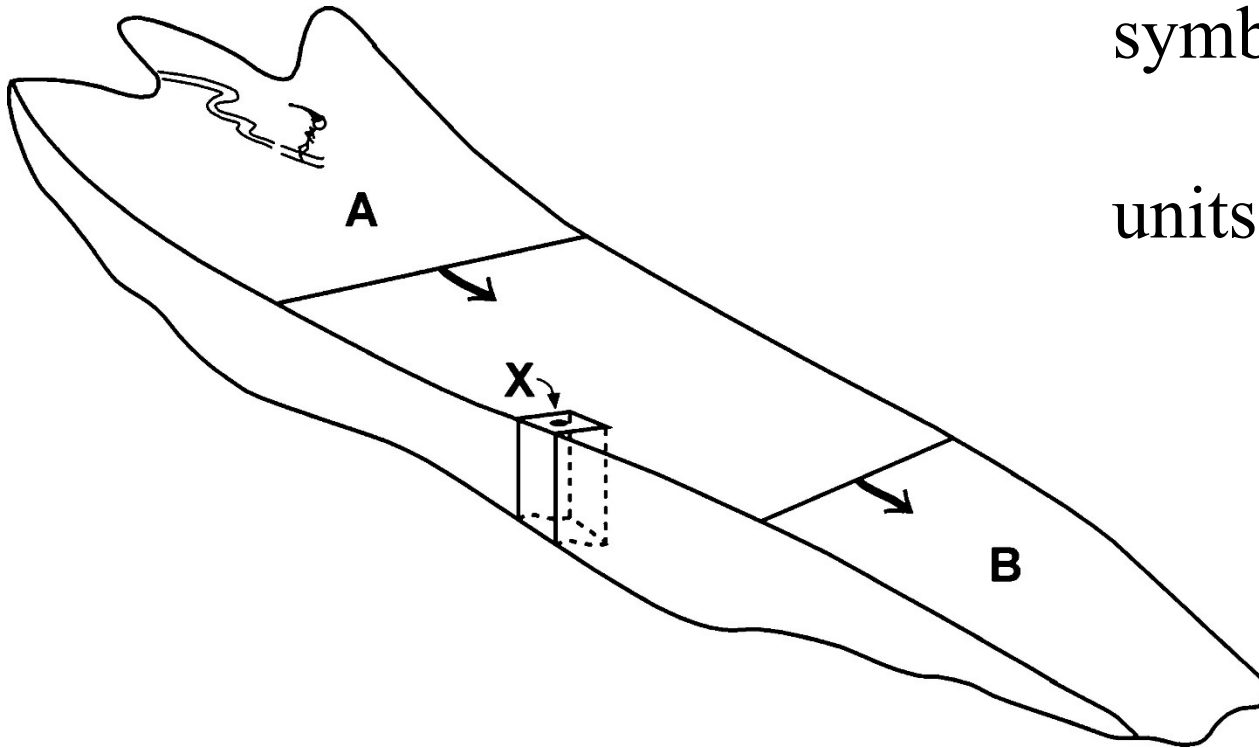
Change in mass per unit area

Includes surface, englacial and basal components.

(not ice flow)

symbol: b

units: kg m^{-2}



Cuffey and Paterson (2010)

Specific mass balance rate

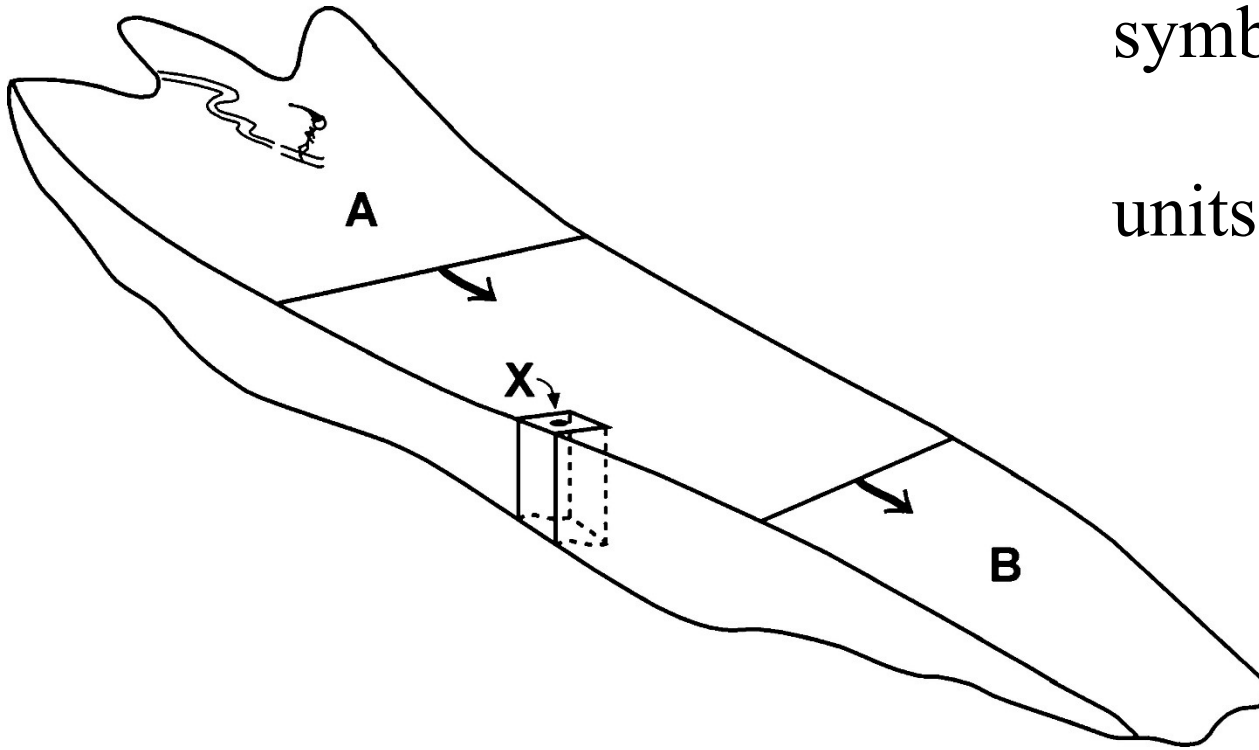
Change in mass per unit area per unit time

Includes surface, englacial and basal components.

(not ice flow)

symbol: $\dot{b}$

units: $\text{kg m}^{-2} \text{a}^{-1}$



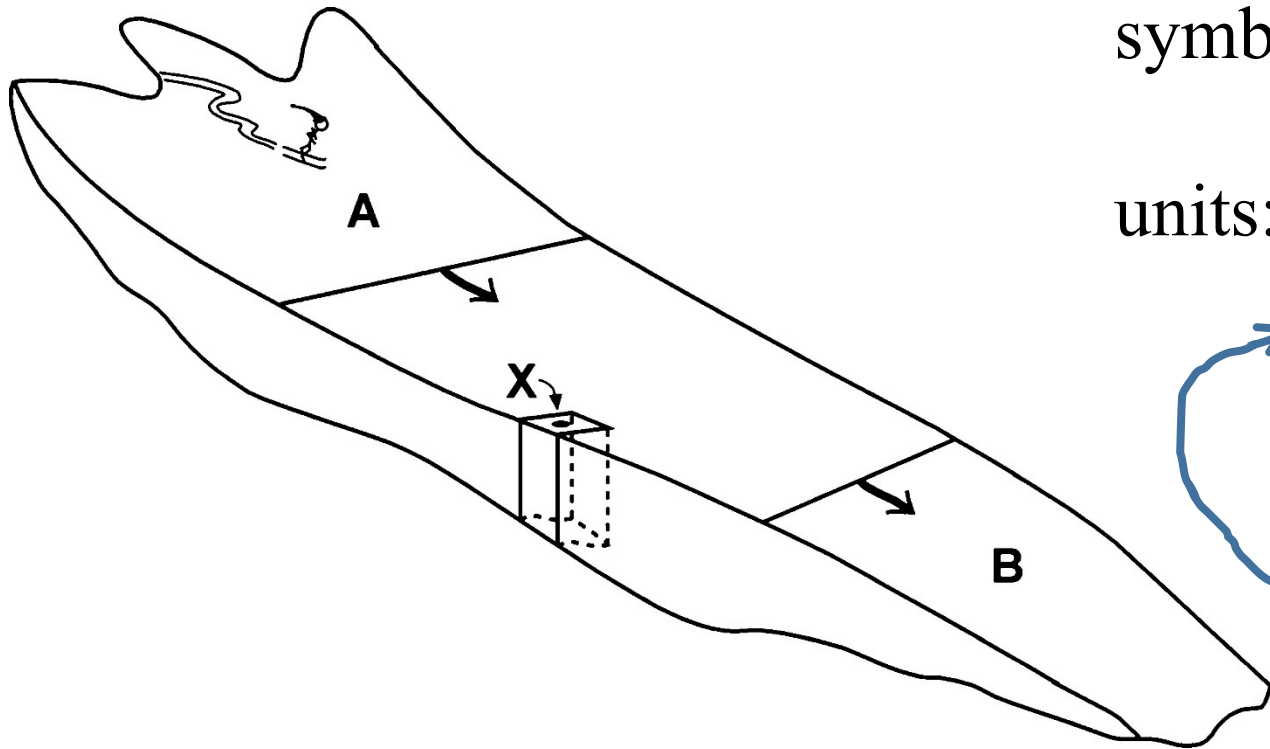
Cuffey and Paterson (2010)

Specific mass balance rate

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symbol: $\dot{b}$

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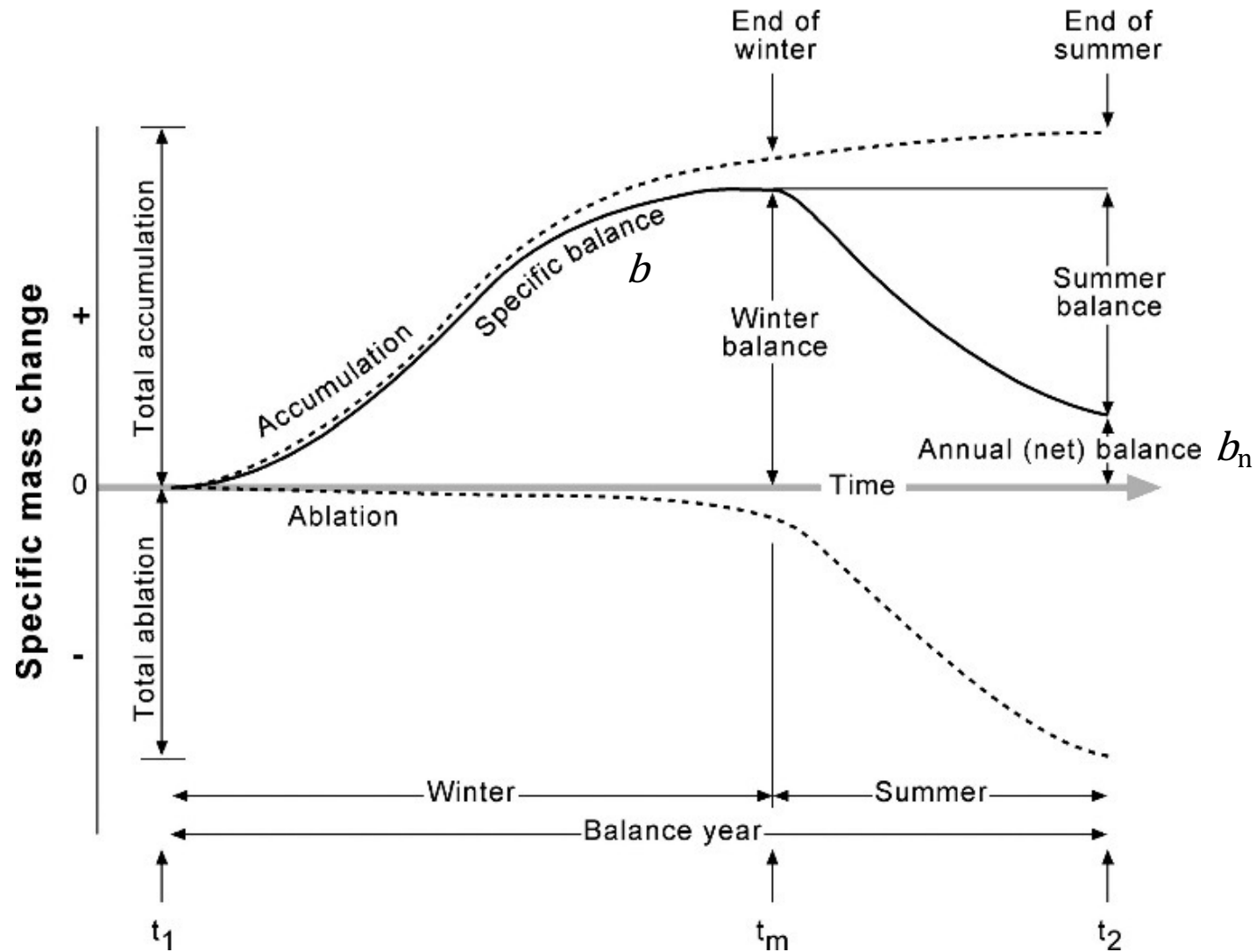
m a^{-1}

m w.e. a^{-1}

This is “ice-equivalent”
used in ice-sheet
modelling.

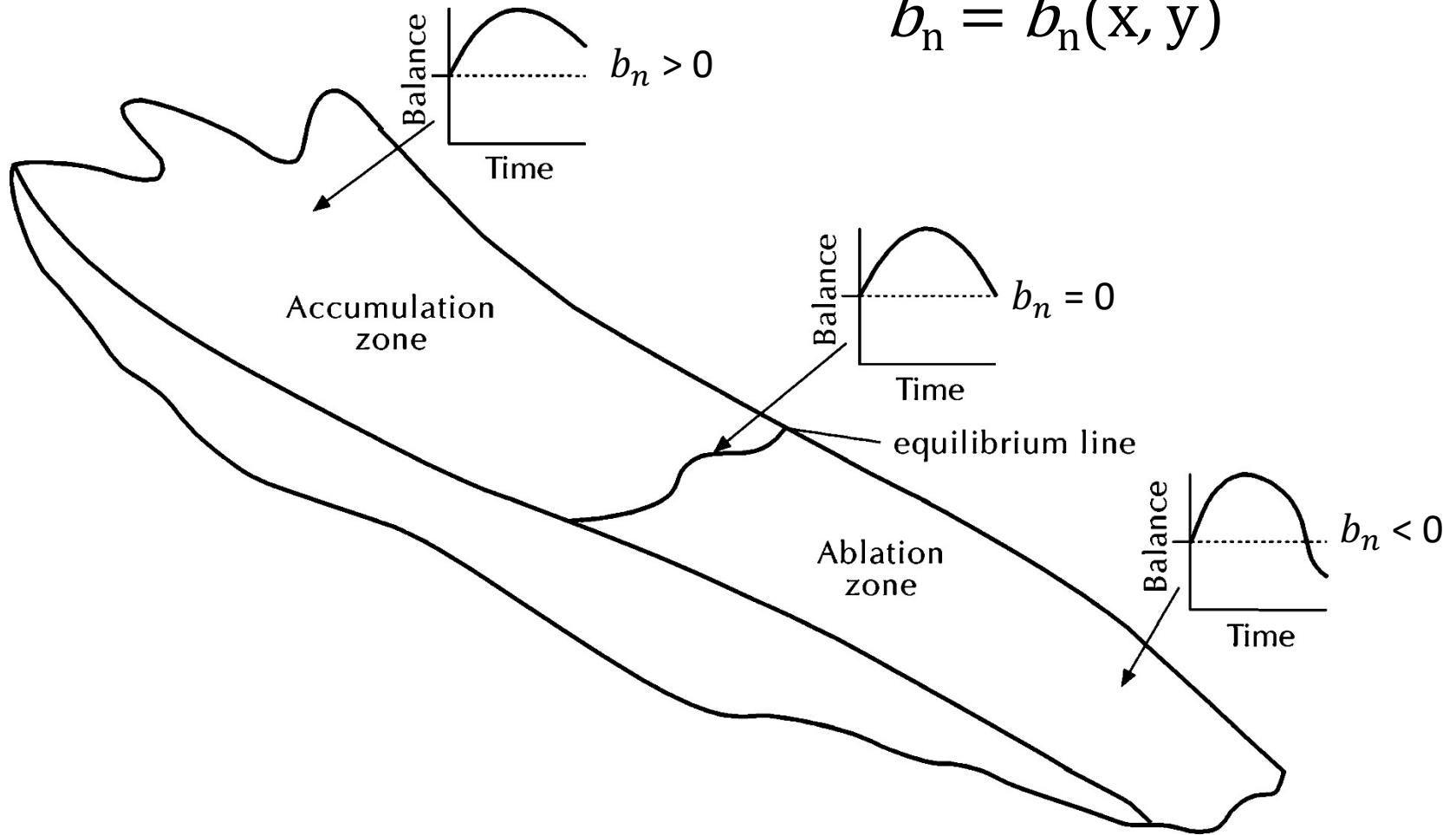
Cuffey and Paterson (2010)

Annual or net specific balance, b_n



Spatially-varying net specific balance

$$b_n = b_n(x, y)$$



Cuffey and Paterson (2010)

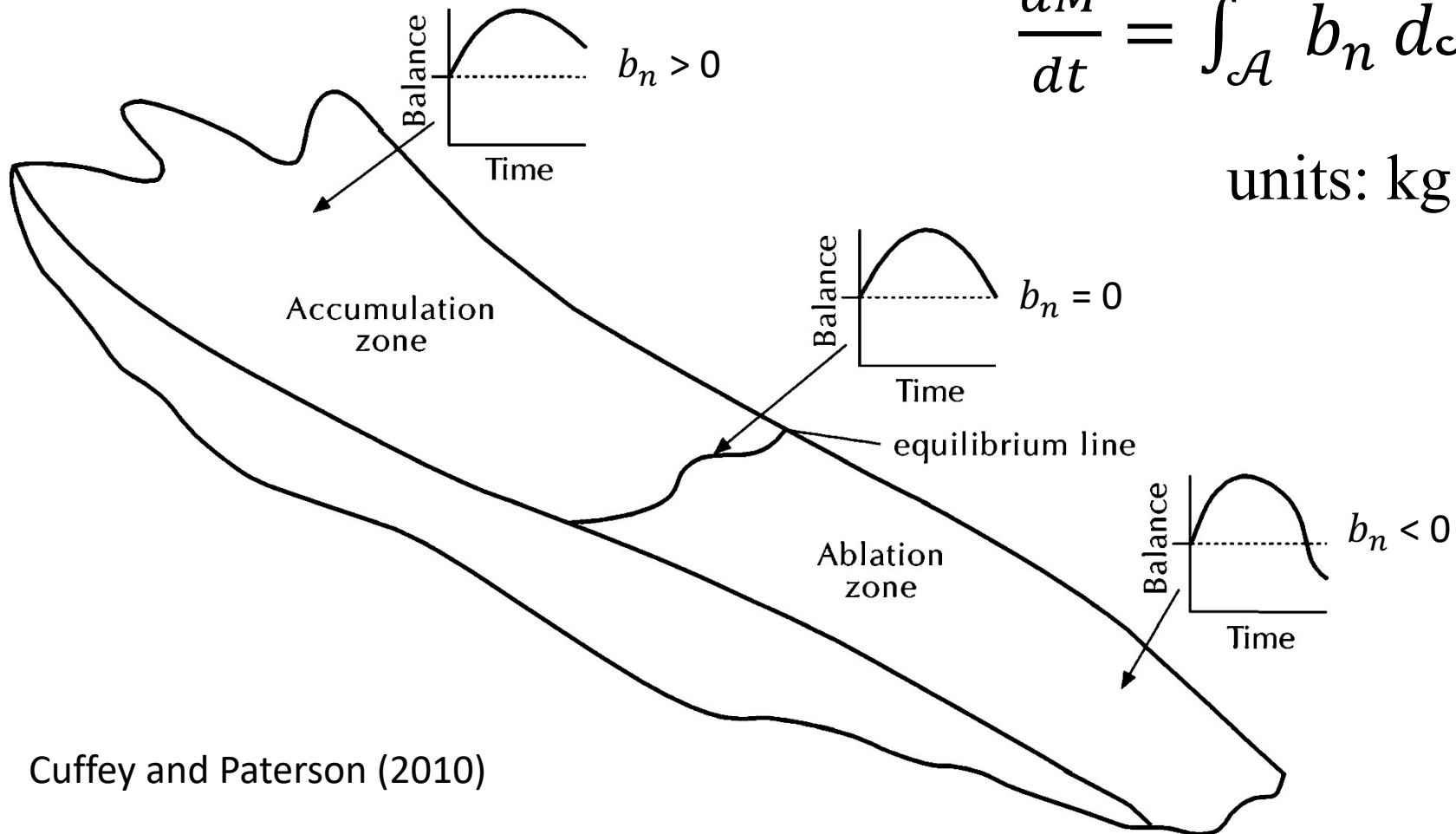
Glacier balance

Change in total mass of the glacier, M , per unit time

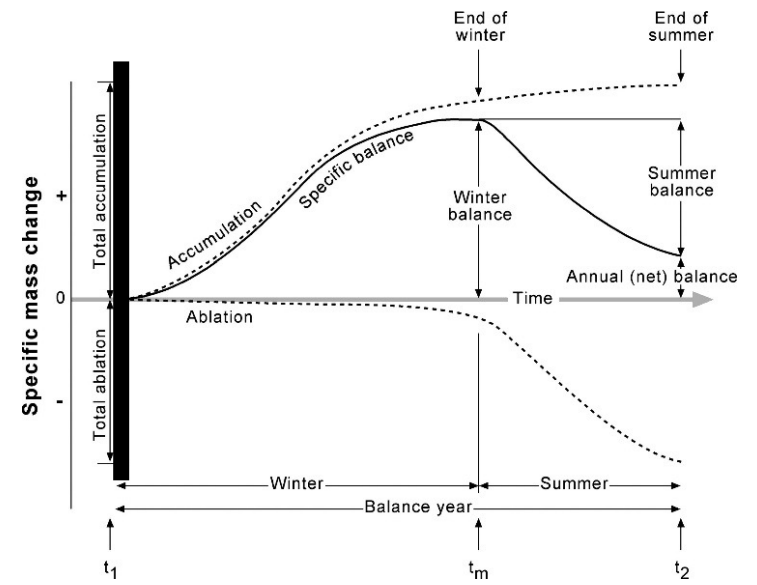
integrate specific balance

$$\frac{dM}{dt} = \int_{\mathcal{A}} b_n d\mathcal{A}$$

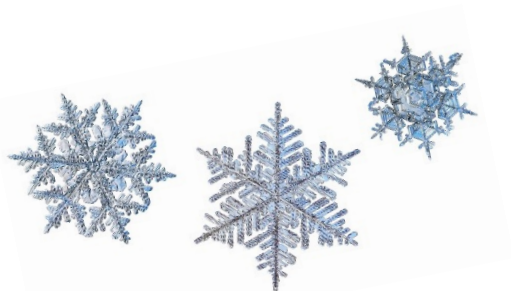
units: kg a^{-1}



Cuffey and Paterson (2010)



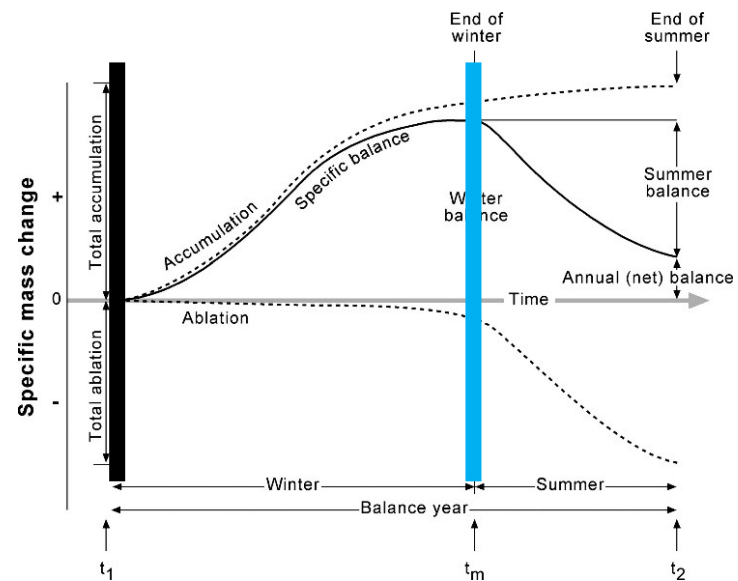
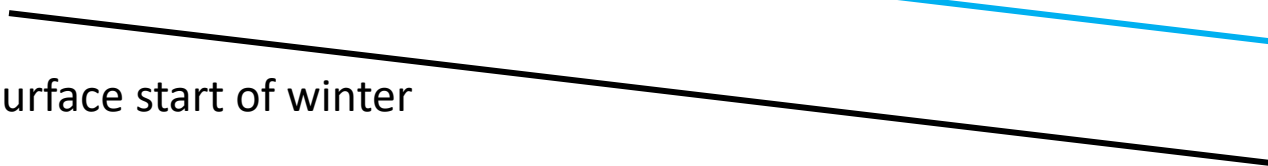
Ice surface start of winter

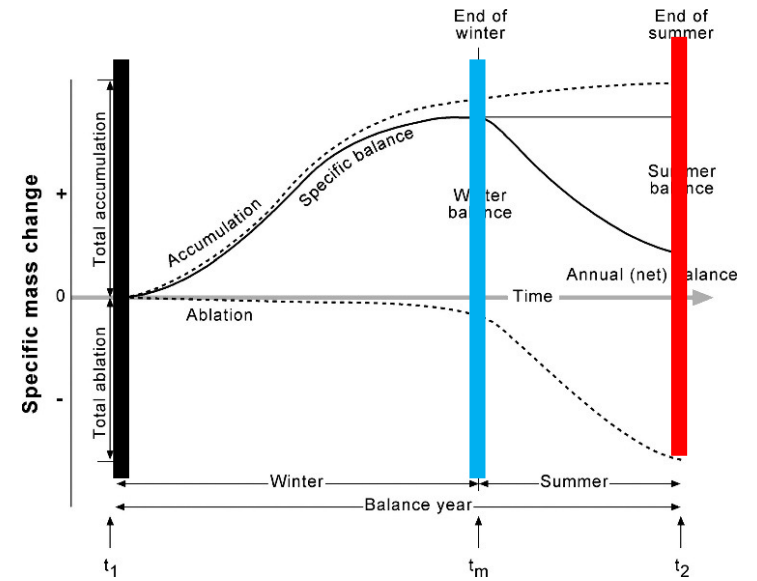
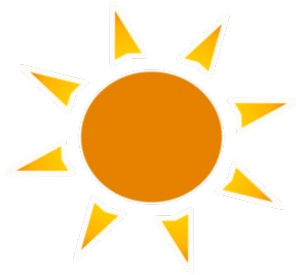


Ice surface end of winter



Ice surface start of winter

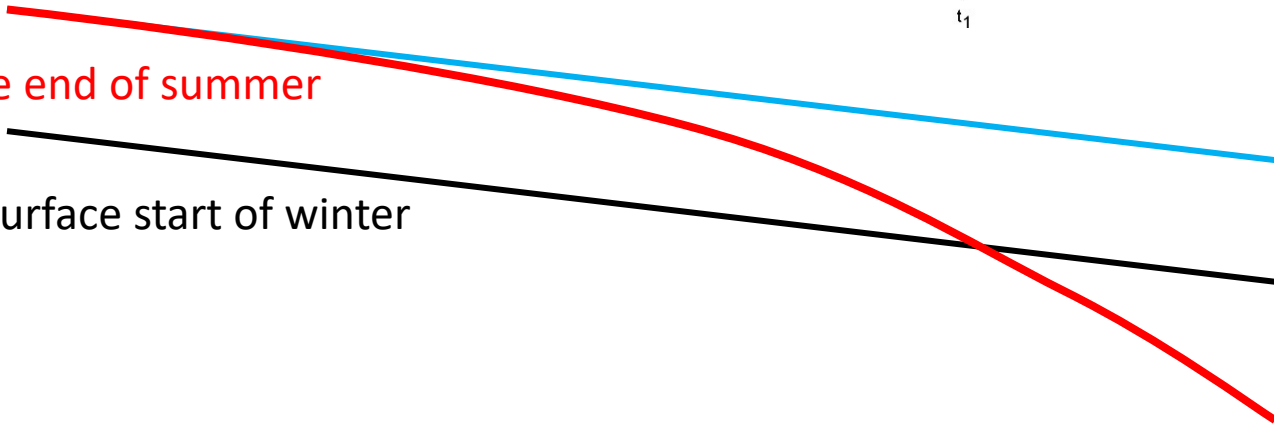


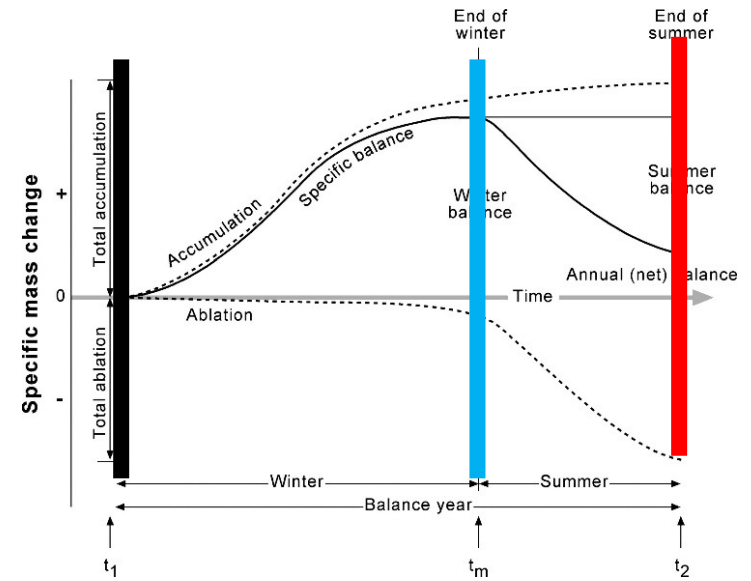
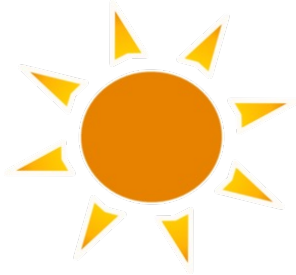


Ice surface end of winter

Ice surface end of summer

Ice surface start of winter

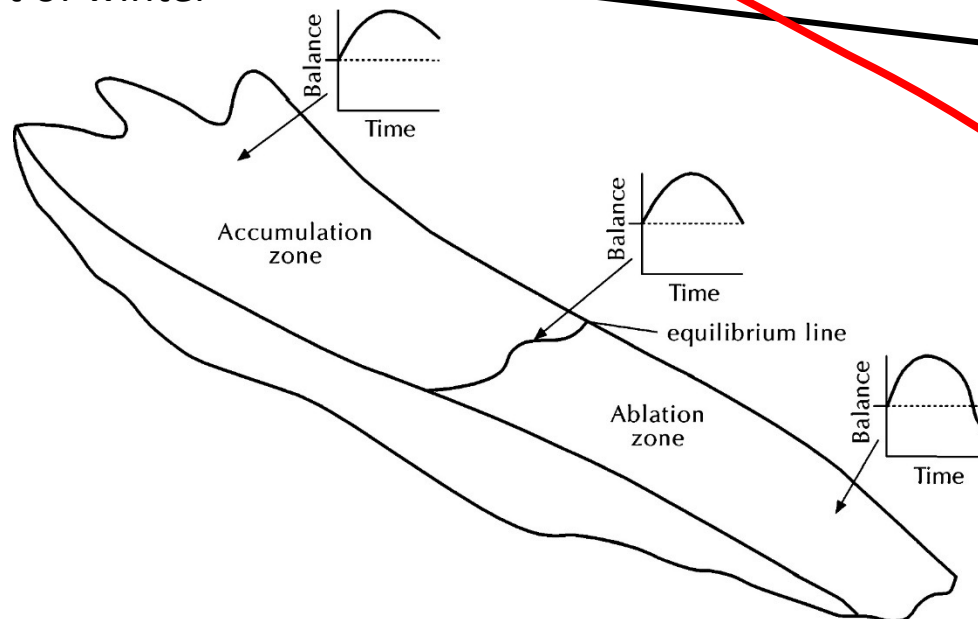




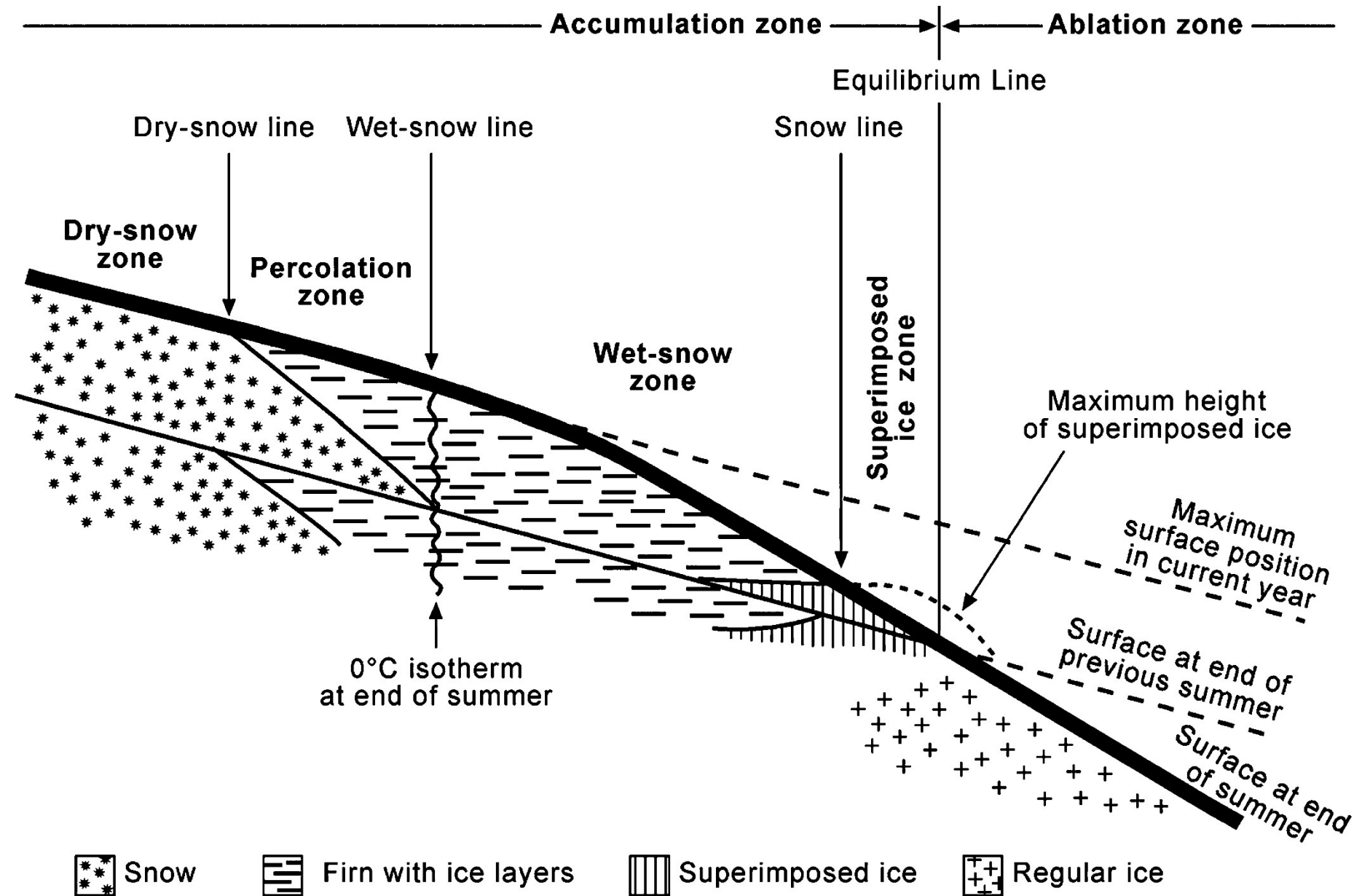
Ice surface end of winter

Ice surface end of summer

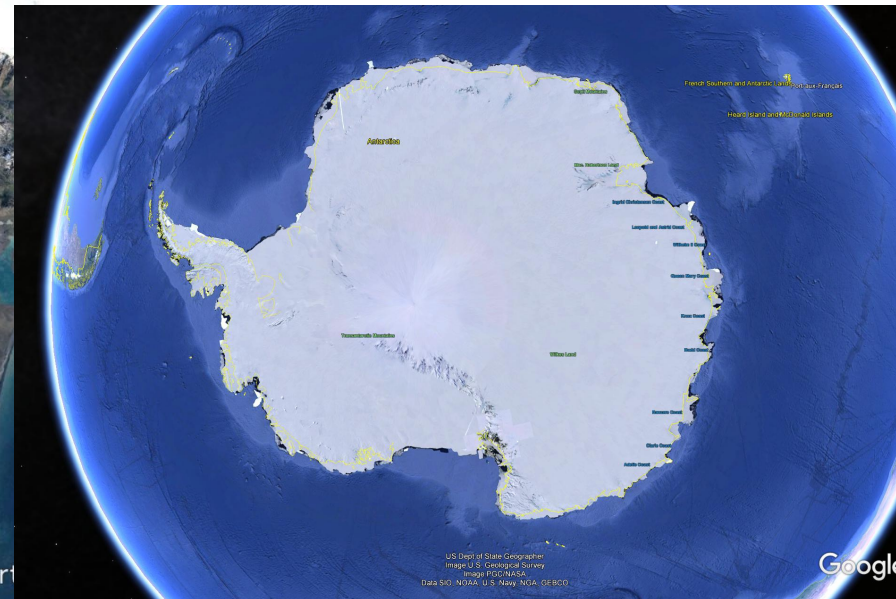
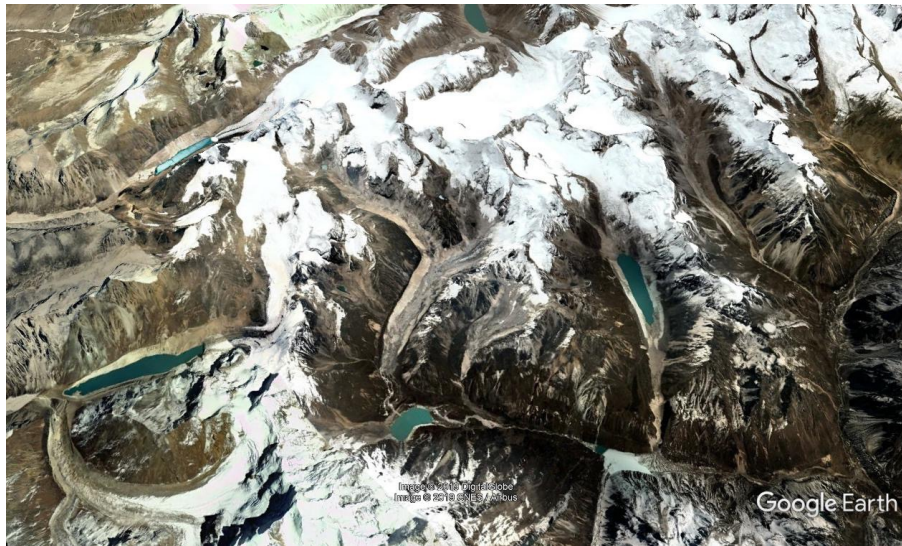
Ice surface start of winter



Surface facies



Accumulation zones from space

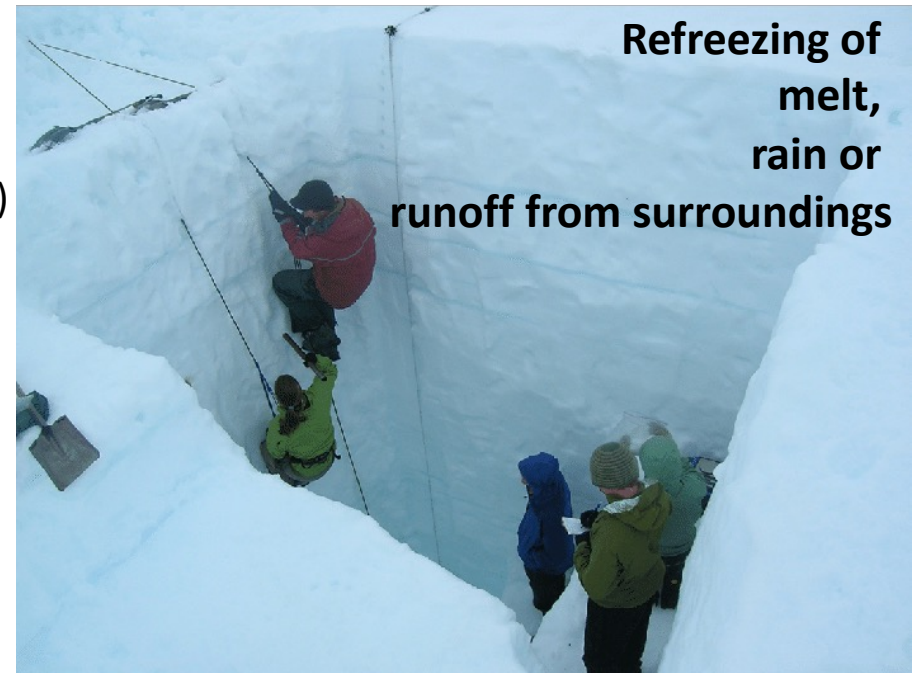


Accumulation processes

- Snow fall (dominates almost everywhere)
- Basal freezing



Avalanching



Deposition of hoar

Greenland's percolation zone





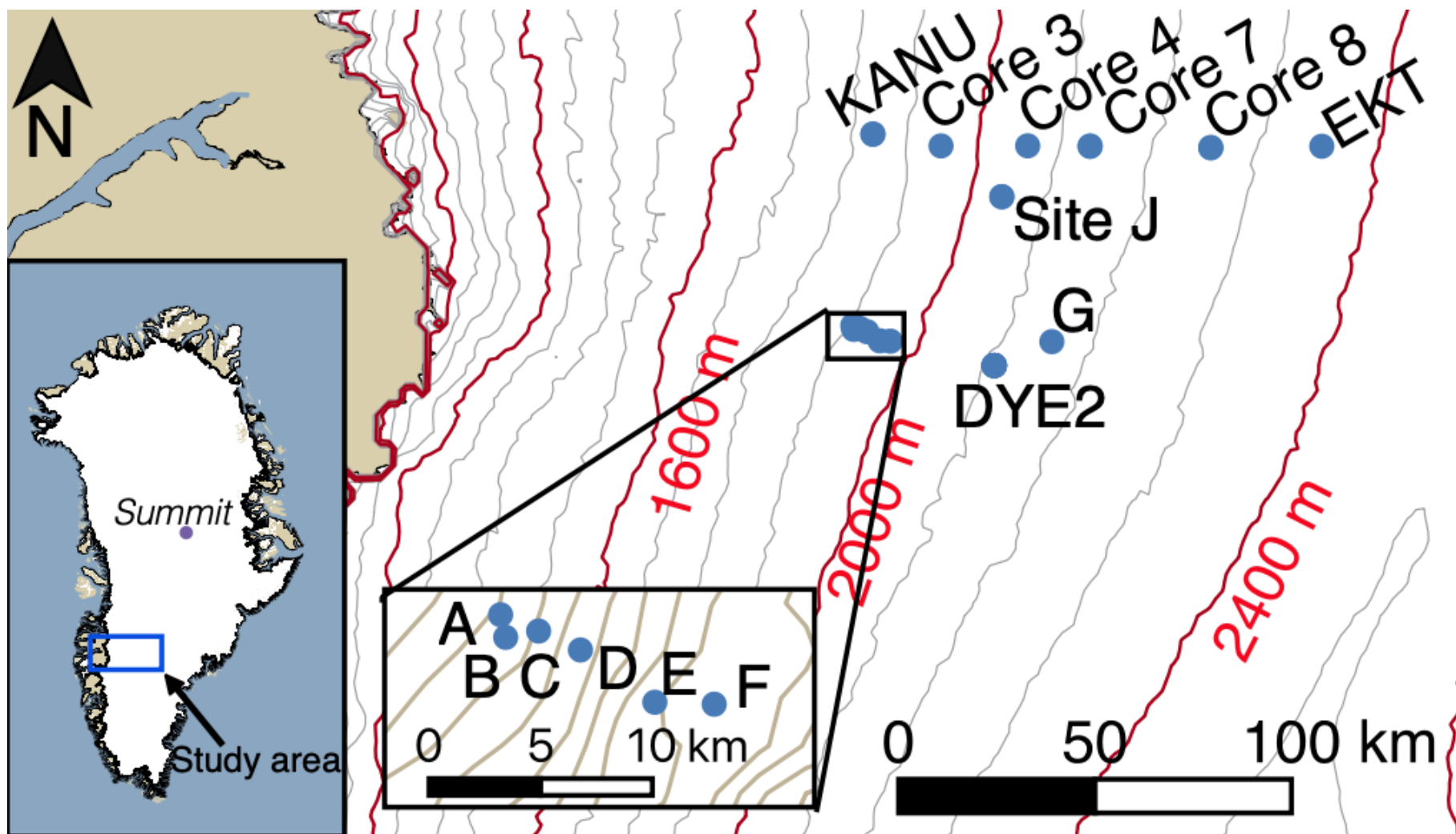


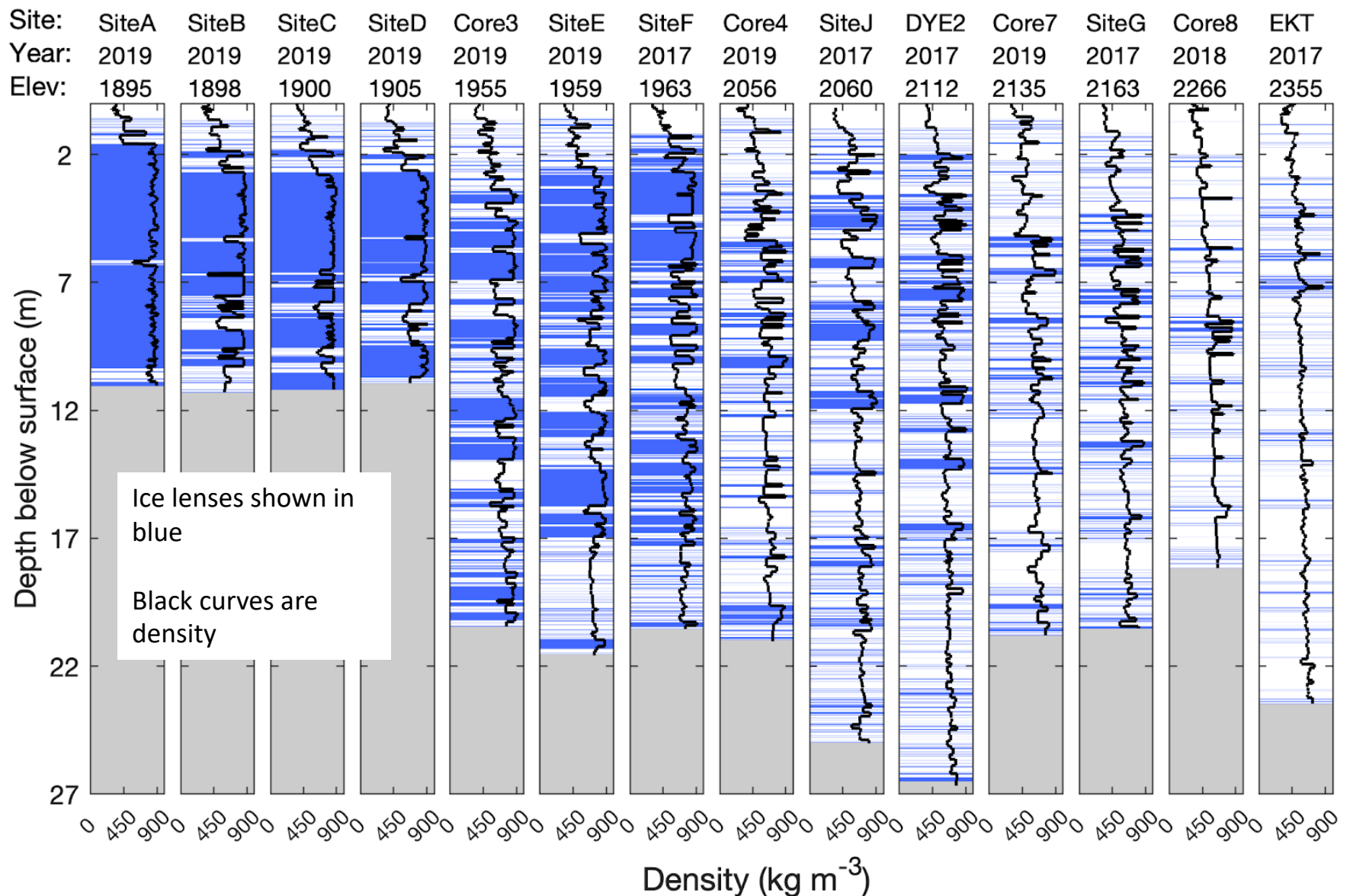


Photo: Regine Hock

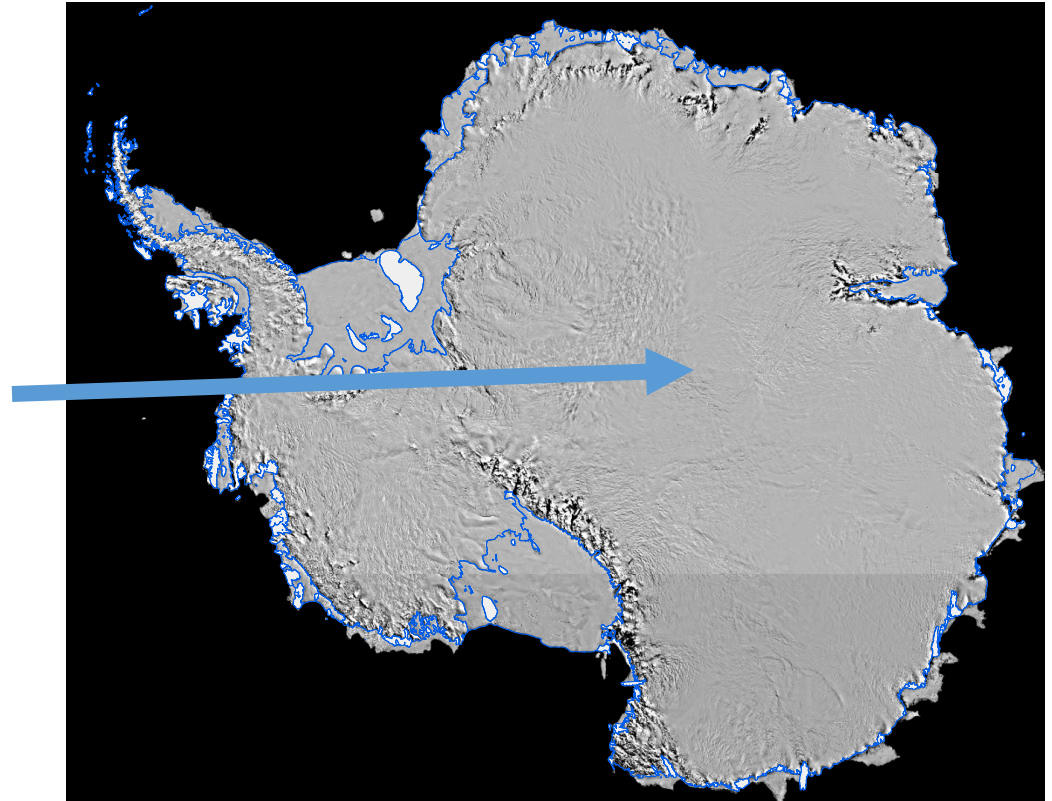
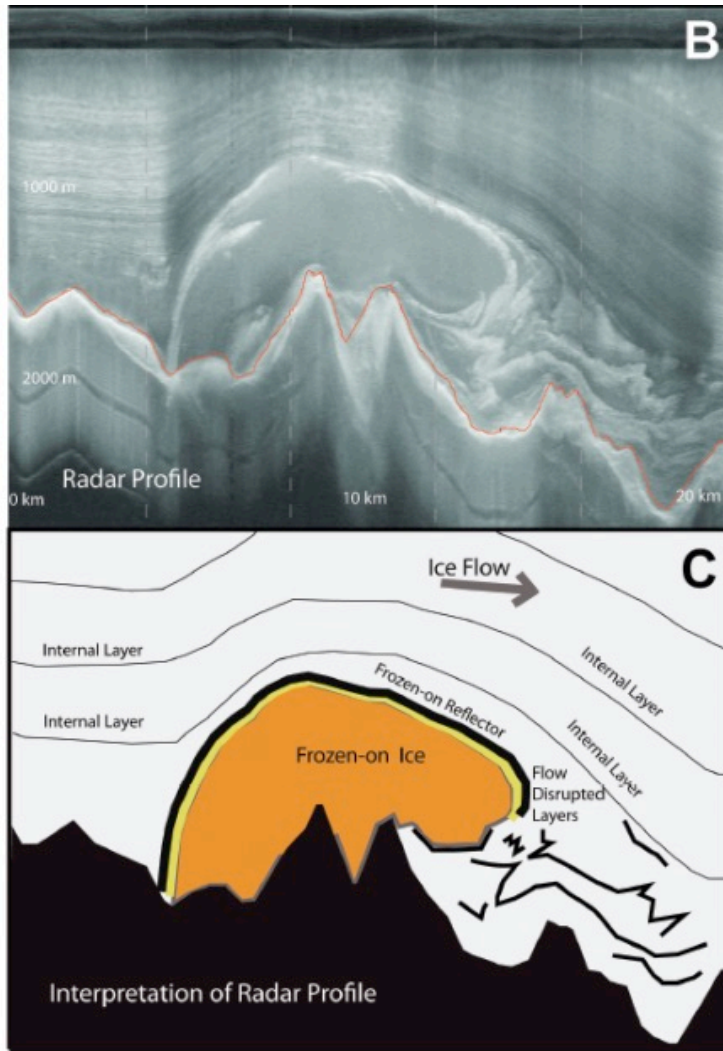
Ice lens



Increasing elevation →

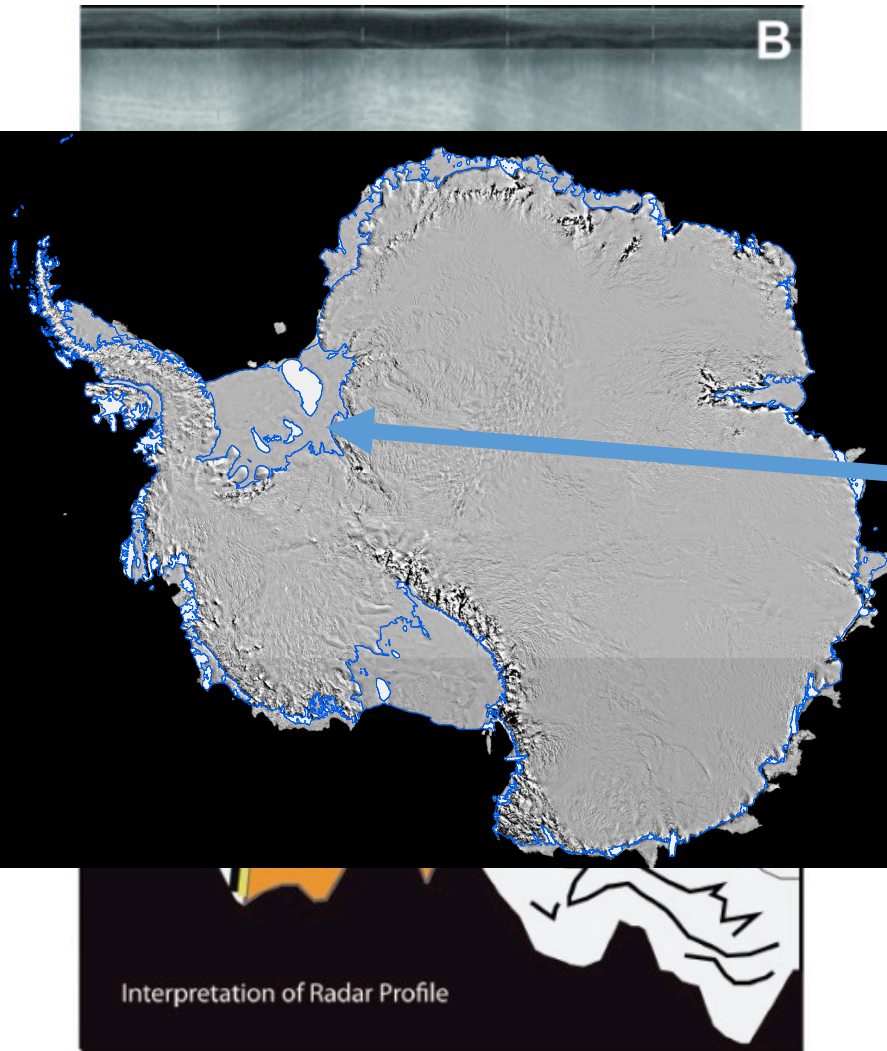


Basal freeze-on on ice shelves and on grounded ice

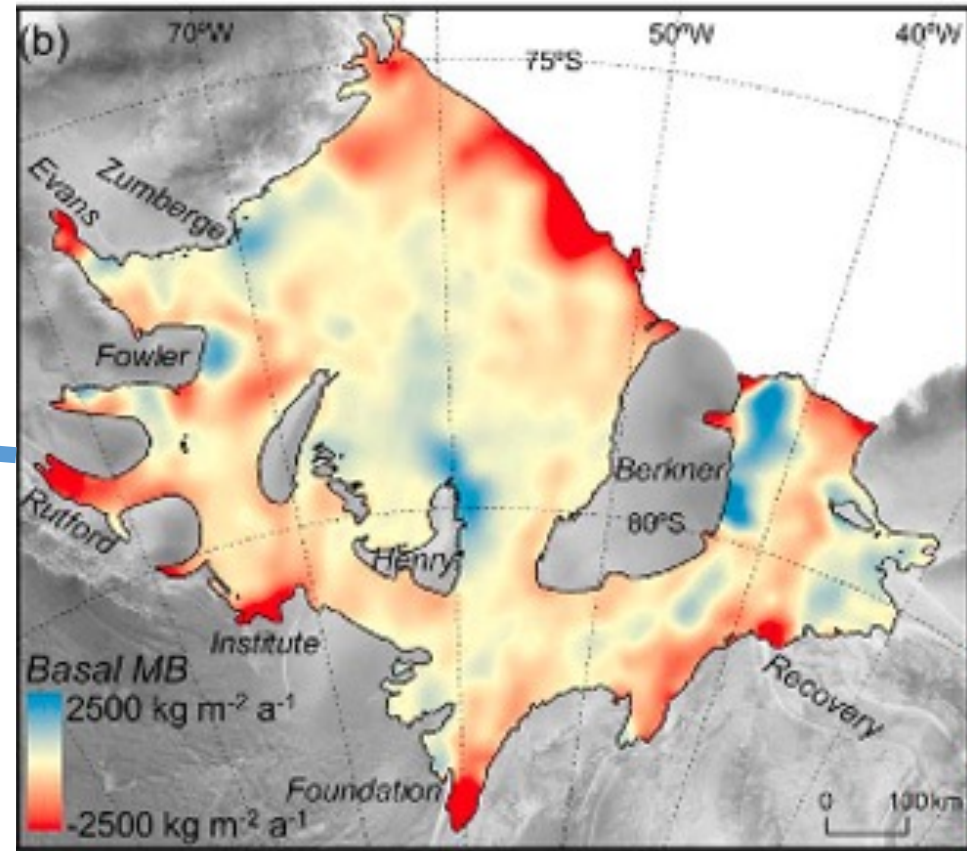


Bell et al. (2011)

Basal freeze-on on ice shelves and on grounded ice



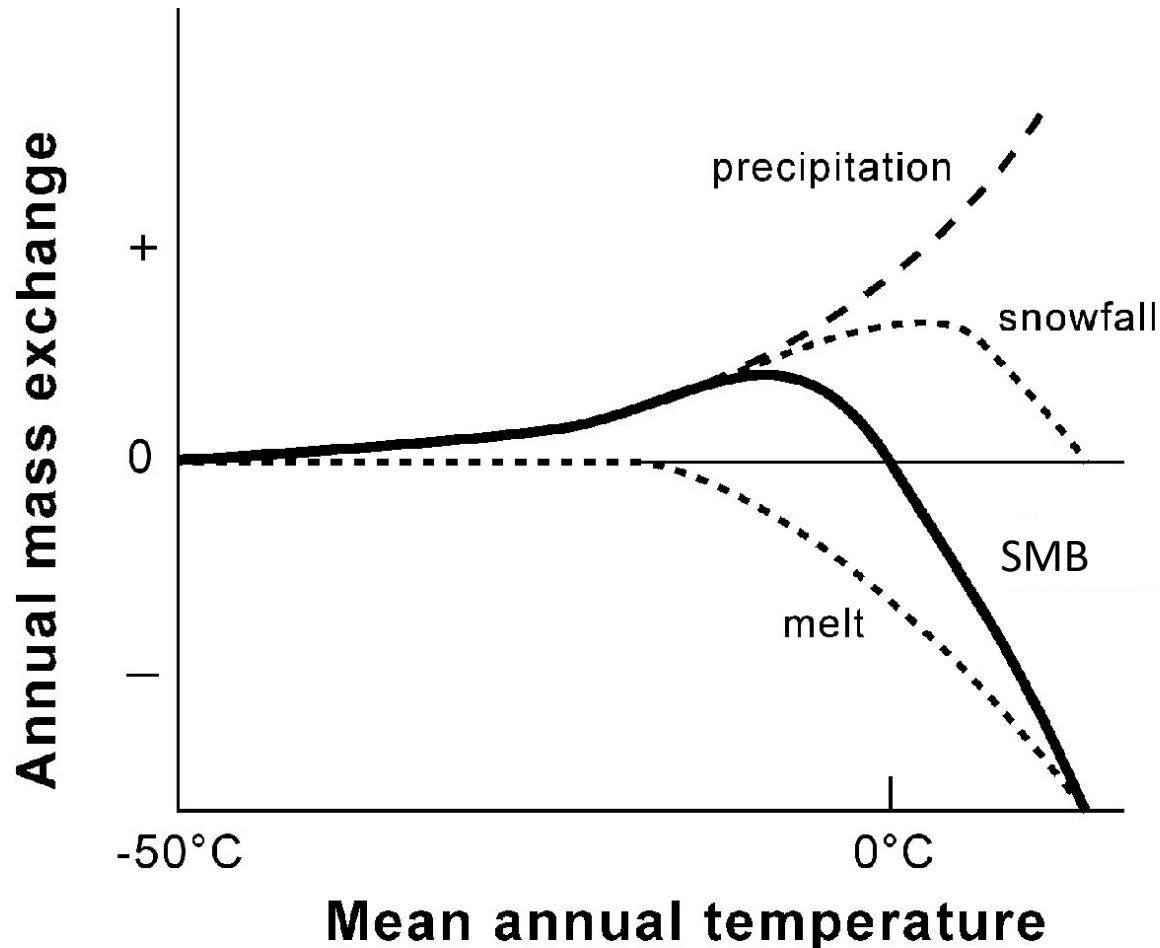
Bell et al. (2011)



Moholdt et al. (2014)

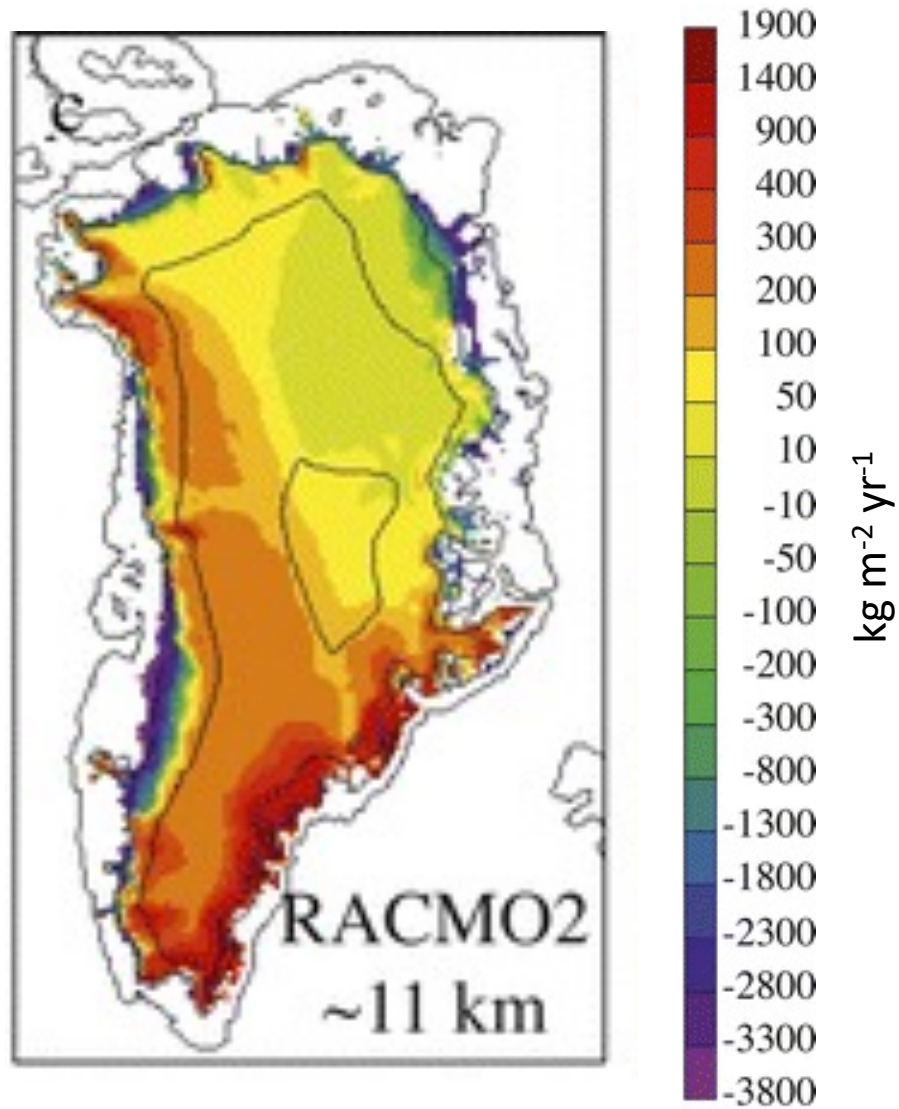
Surface mass balance (SMB) often dominates mass balance (MB)

- Temperature is the primary control on SMB



Long-term SMB of Greenland

Vizcaíno et al. (2014)



2 minute break



Firn

-- snow that has survived on the surface of a glacier for a full melt season.

Snow starts out low density and increases in density over time until it becomes ice.

Firn densification

Q: Why is it important to understand it?

Q: What controls densification?



Maximum pressure in a snowball?



Roll over image to zoom in



CAMRY

Camry Digital Hand Dynamometer Grip Strength Measurement Meter Auto Capturing Hand Grip Power 200 Lbs / 90 Kgs



192 customer reviews | 24 answered questions

Amazon's Choice for "dynamometer"

Price: **\$29.99** & **FREE Shipping**. [Details](#)

In Stock.

Want it Monday, Jan. 22? Order within **10 hrs 27 mins** and choose **Standard Shipping** at checkout. [Details](#)

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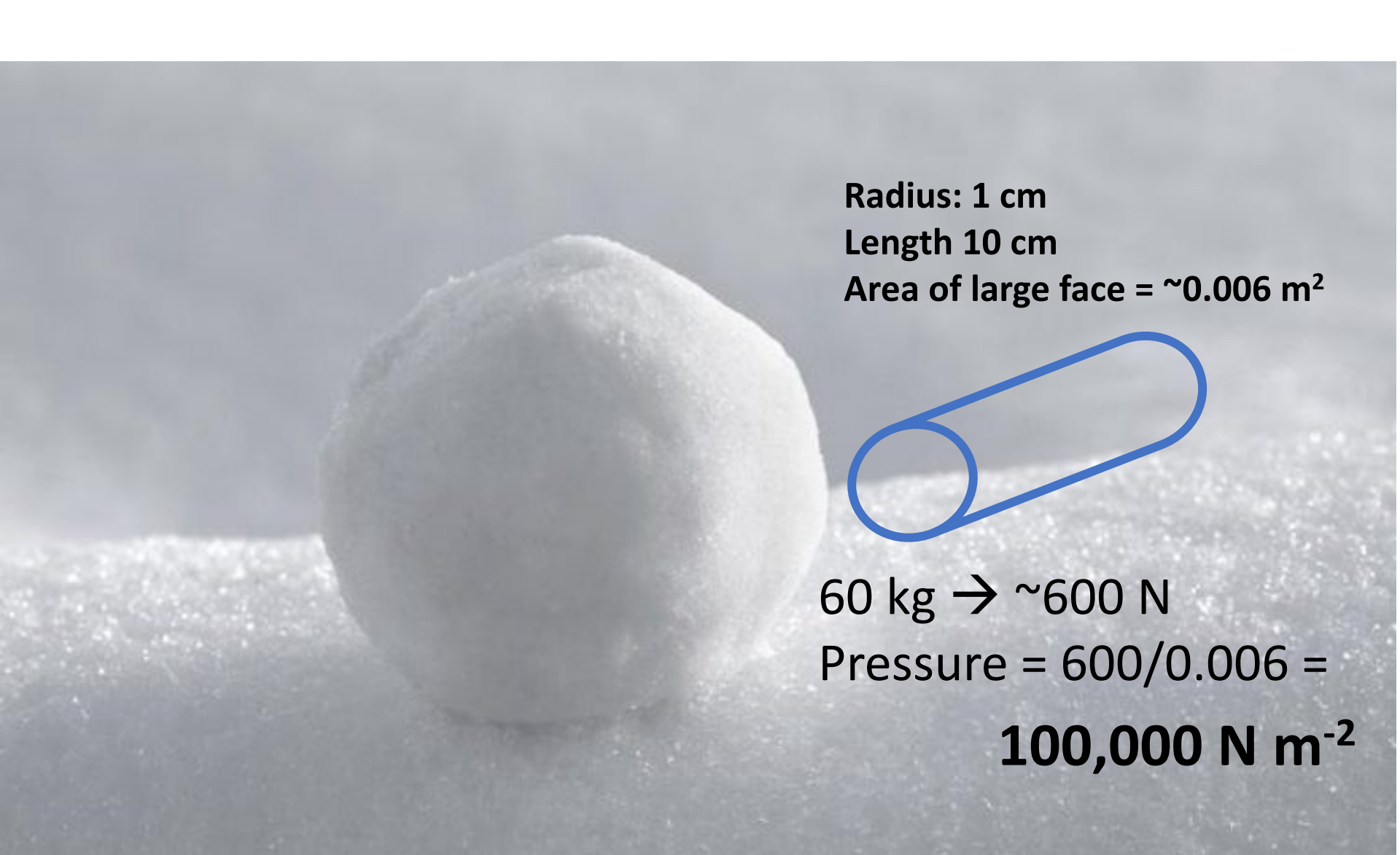
- **ACCURACY** - Equipped with high precision strain gauge sensor, the hand dynamometer gives you accurate momentary digital reading of gripping power. Measuring Capacity: 198lbs / 90kgs; Division: 0.2lbs / 0.1kgs
- **USER FRIENDLY** - Squeeze the hand dynamometer with maximum isometric effort for at least 5 seconds. After testing, the LCD will automatically display the maximum grip value and a grip value status bar showing the status of "weak", "normal" or "strong" according to age and gender preset for each test.
- **MEMORY RECORD** - The device can store data for up to 19 definable users, and their records can be recalled anytime when needed; Moreover, the incremental increase or decrease from the last record will be automatically shown for comparison after each test.
- **Adjustability** - Turn the adjustable gear to get an ideal grasp according to the size of hands. It is good fit for both big hand and small hand.
- **EASY TO READ** - Large LCD screen shows all measurements clearly on the same screen. Displayed measurements include: user code, age, gender, grip value in pound or kg and grip status. Engineered for home, sports and clinical use. 2* AAA batteries included; 5-year product warranty.

Used & new (2) from **\$29.99** & **FREE shipping**. [Details](#)

Maximum pressure in a snowball?

AGE	Weak	Normal	Strong
10-11	< 12.6	12.6-22.4	> 22.4
12-13	< 19.4	19.4-31.2	> 31.2
14-15	< 28.5	28.5-44.3	> 44.3
16-17	< 32.6	32.6-52.4	> 52.4
18-19	< 35.7	35.7-55.5	> 55.5
20-24	< 36.8	36.8-56.6	> 56.6
25-29	< 37.7	37.7-57.5	> 57.5
30-34	< 36.0	36.0-55.8	> 55.8
35-39	< 35.8	35.8-55.6	> 55.6
40-44	< 35.5	35.5-55.3	> 55.3
45-49	< 34.7	34.7-54.5	> 54.5
50-54	< 32.9	32.9-50.7	> 50.7
55-59	< 30.7	30.7-48.5	> 48.5
60-64	< 30.2	30.2-48.0	> 48.0
65-69	< 28.2	28.2-44.0	> 44.0
70-99	< 21.3	21.3-35.1	> 35.1

Typical grip
strength: 30-60 kg



Radius: 1 cm

Length 10 cm

Area of large face = $\sim 0.006 \text{ m}^2$

$60 \text{ kg} \rightarrow \sim 600 \text{ N}$

Pressure = $600 / 0.006 =$

$100,000 \text{ N m}^{-2}$

$50,000 - 100,000 \text{ N m}^{-2}$

$50 - 100 \text{ kPa}$

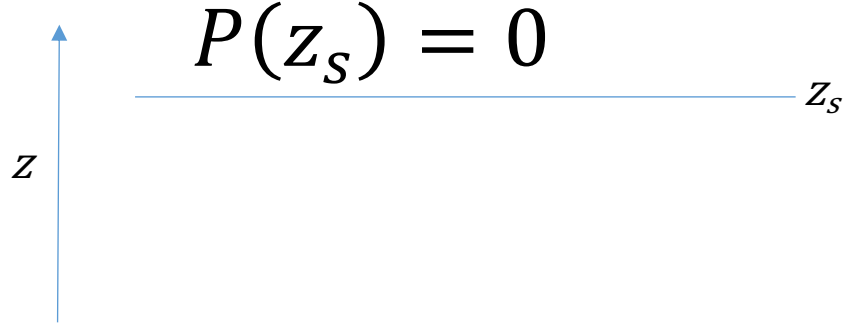
Overburden pressure

We want an equation for how pressure P varies with depth z

Overburden pressure

We want an equation for how pressure P varies with depth z

$$\frac{dP}{dz} = -\rho(z)g$$

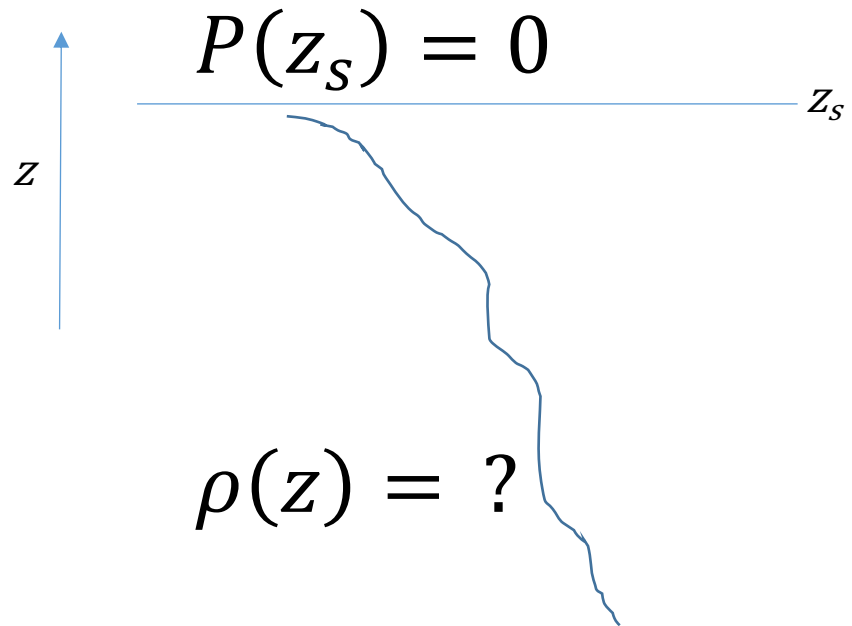


A diagram illustrating the coordinate system for overburden pressure. A vertical blue arrow points upwards and is labeled with the variable z . To the right of the arrow, a horizontal blue line is drawn, labeled z_s at its right end. Above this horizontal line, the equation $P(z_s) = 0$ is written.

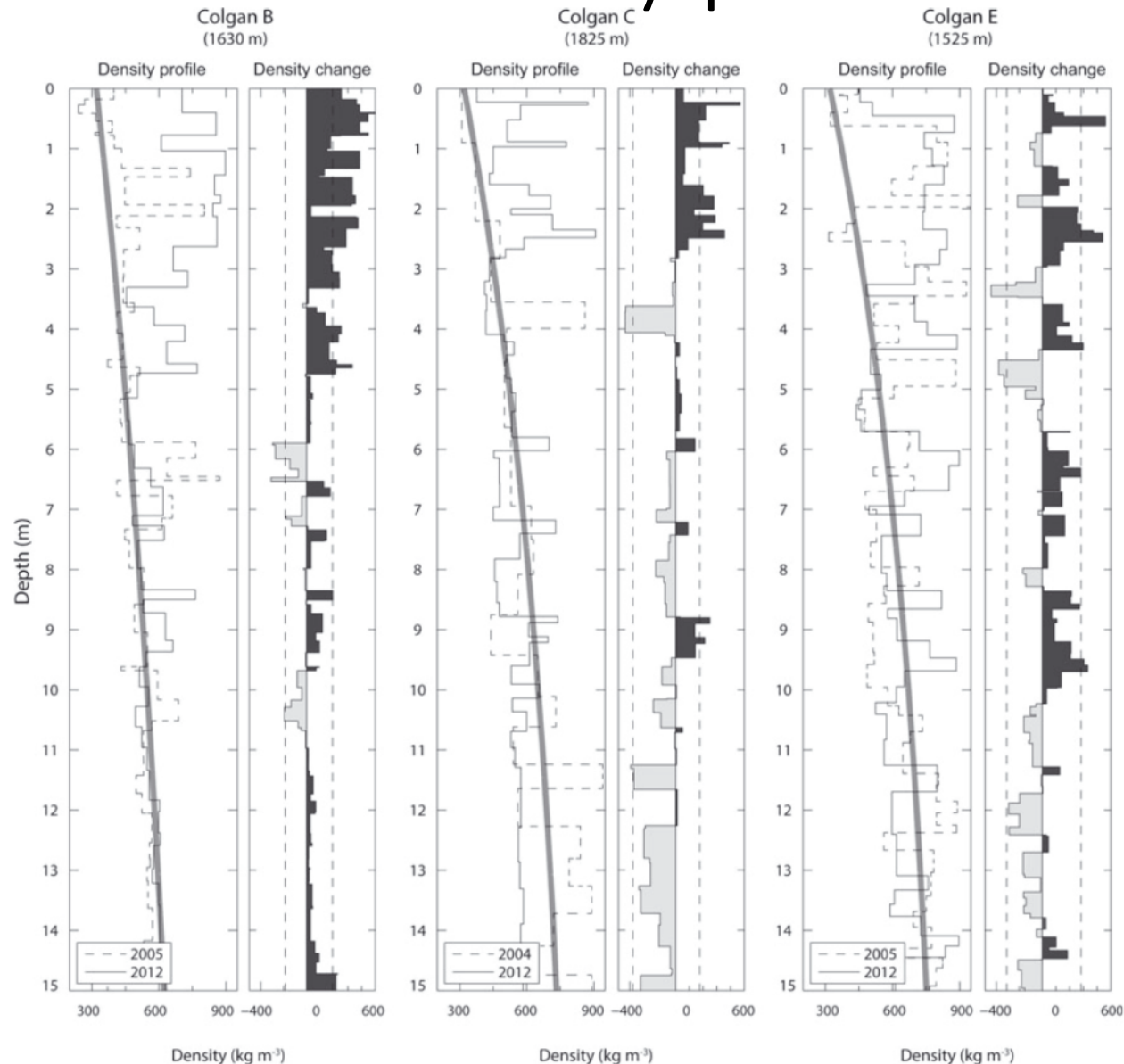
Overburden pressure

We want an equation for how pressure P varies with depth z

$$\frac{dP}{dz} = -\rho(z)g$$



Measured firn density profiles



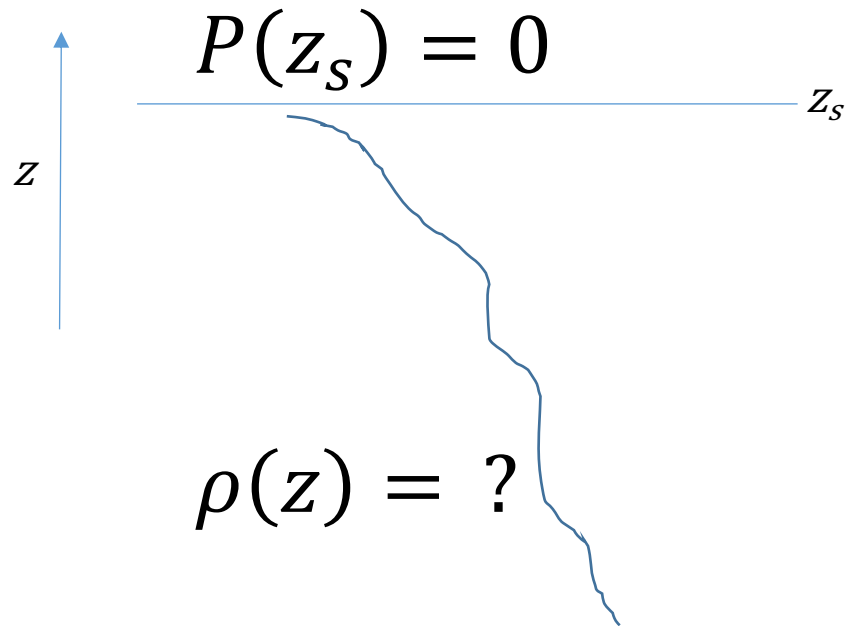
Bezeau et al. (2013)

Fig. 5. Measured and empirically derived (thick gray line) firn density profiles for Colgan B, C and E sites. The firn density change is the 2012 firn density less the 2004/05 firn density. The dashed lines represent ± 2 standard deviations from the mean change of the bottom 10 m.

Overburden pressure

We want an equation for how pressure P varies with depth z

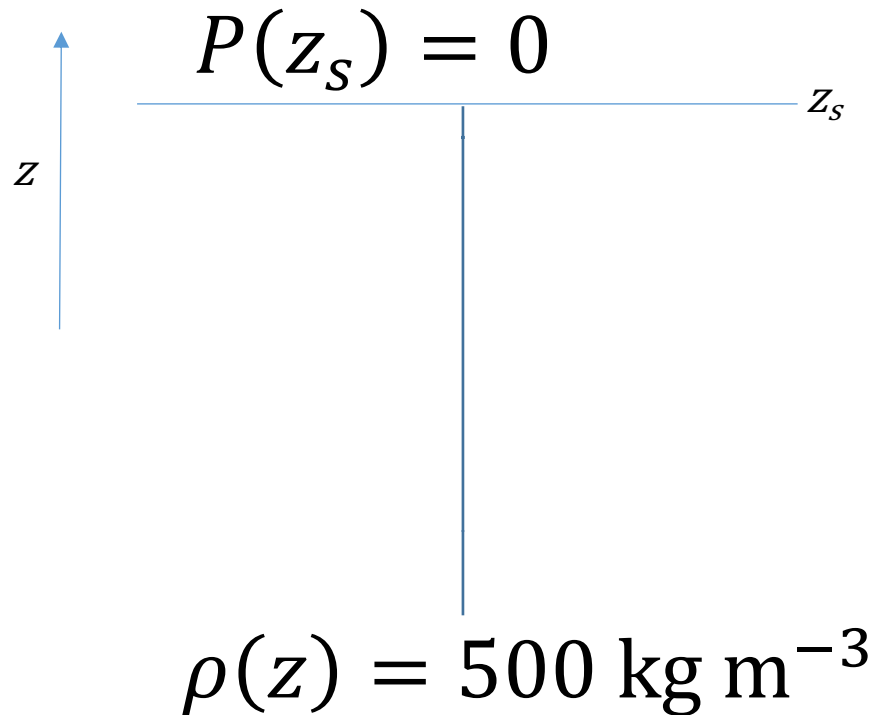
$$\frac{dP}{dz} = -\rho(z)g$$



Overburden pressure

We want an equation for how pressure P varies with depth z

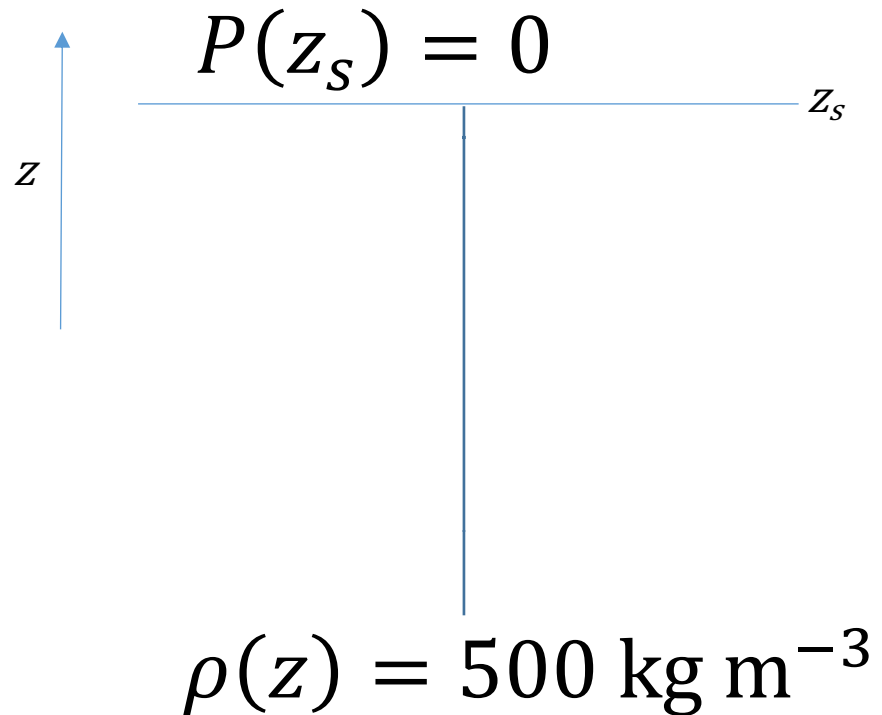
$$\frac{dP}{dz} = -\rho(z)g$$



Overburden pressure

We want an equation for how pressure P varies with depth z

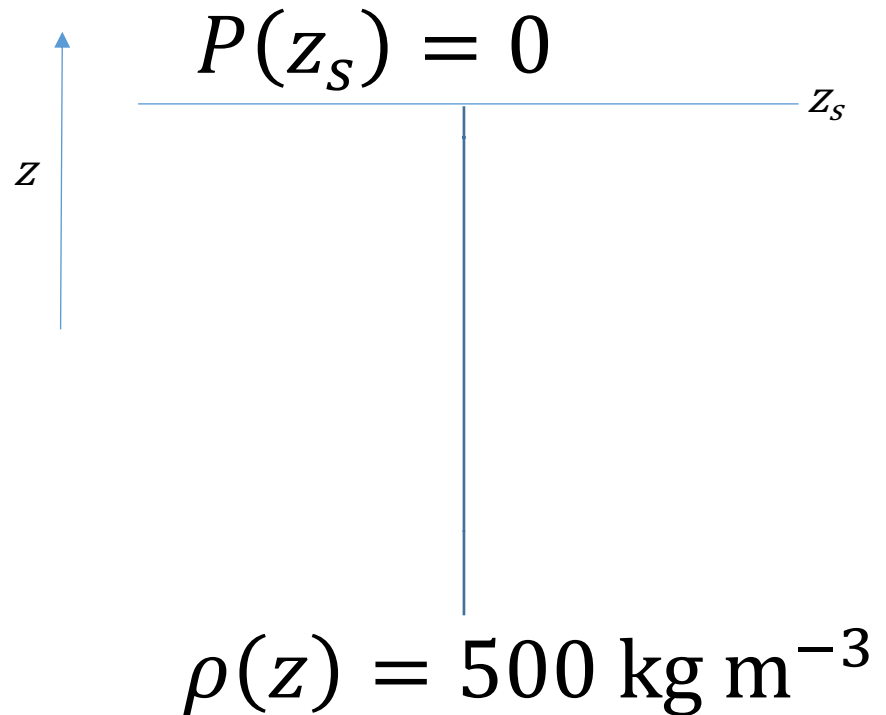
$$P(z) = g\rho(z_s - z)$$



Overburden pressure

We want an equation for how pressure P varies with depth z

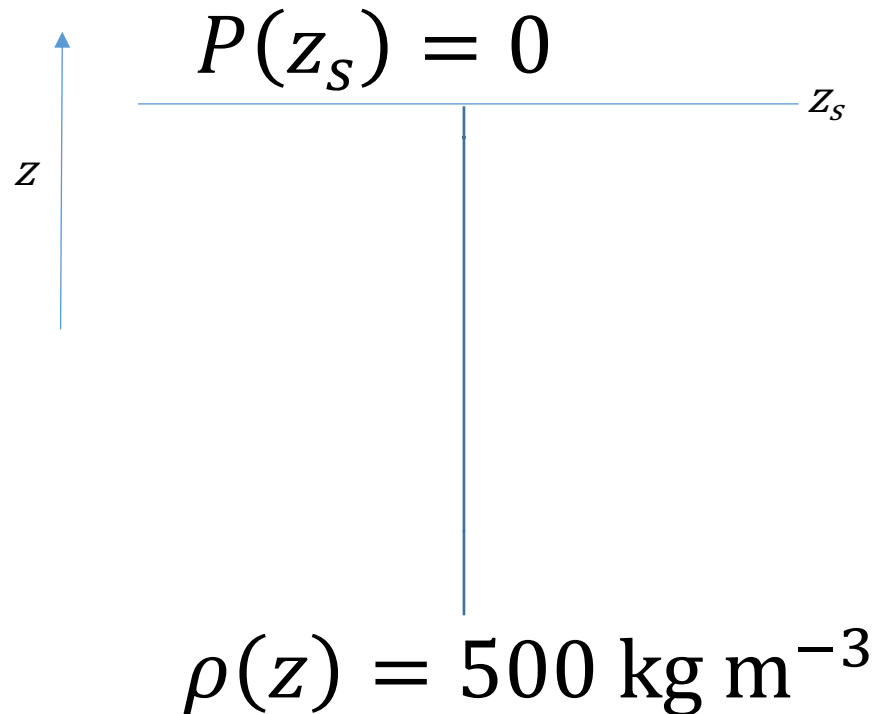
$$\frac{P}{g\rho} = (z_s - z)$$



Overburden pressure

We want an equation for how pressure P varies with depth z

We can exert as much pressure with our hand as 10 - 20 m of firn!



Firn densification processes

Its complicated!

Settling

movement of grains relative to each other
increases density up to $\sim 550 \text{ kg m}^{-3}$

Sublimation

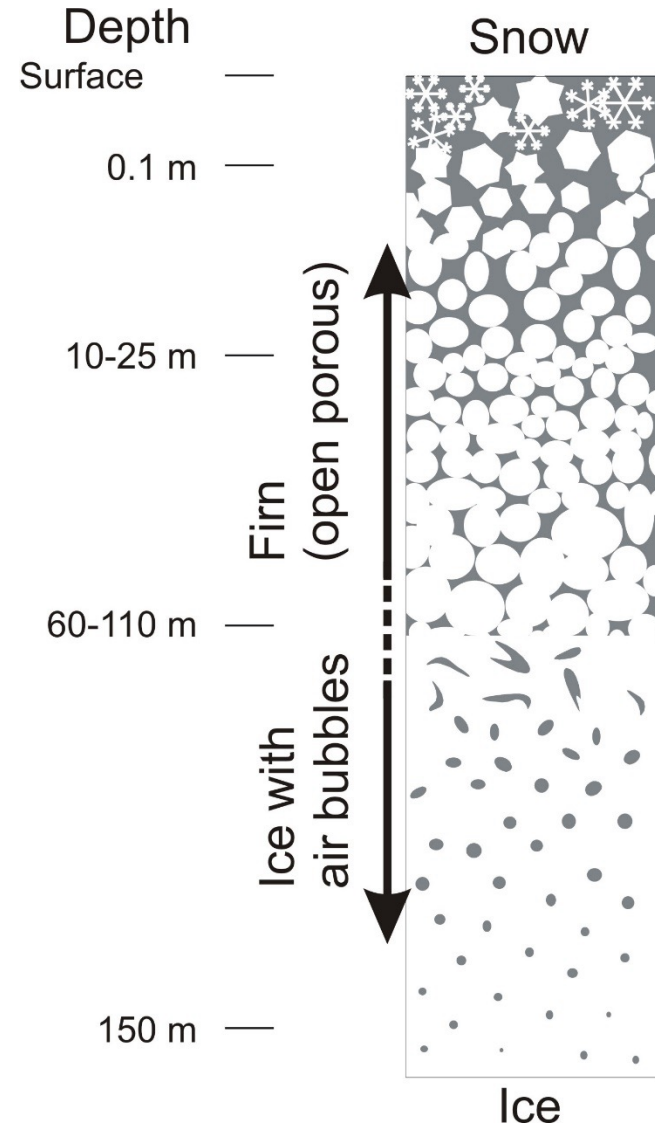
$> 550 \text{ kg m}^{-3}$,
movement of molecules to 'necks' of grains due
to vapour-pressure gradients in pores.

Deformation of grains

Through diffusion creep
(more details next week)

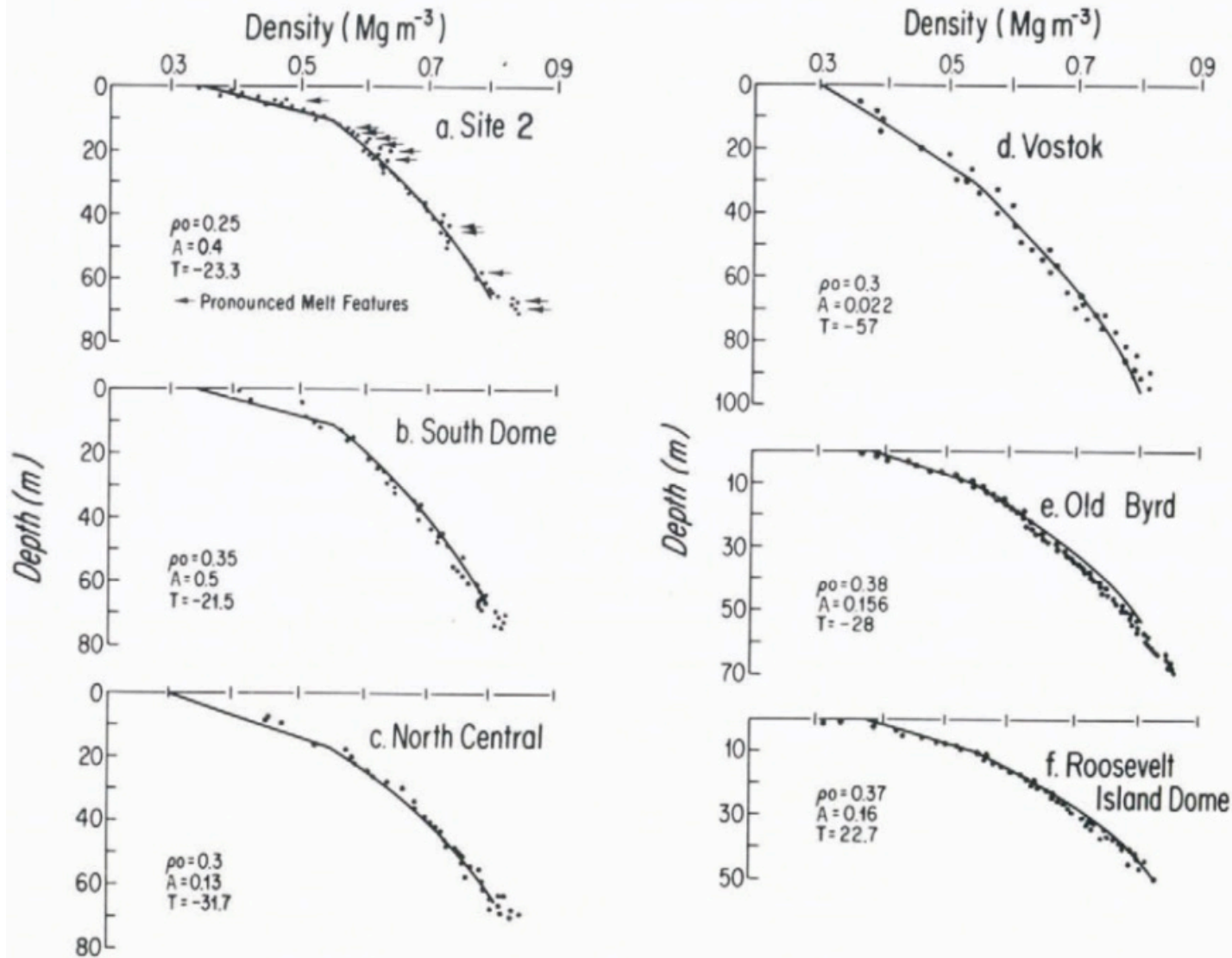
At $\sim 830 \text{ kg m}^{-3}$ pores "close off"
and firn becomes ice.

Compression of air in bubbles allows ice density to
reach $\sim 917 \text{ kg m}^{-3}$.



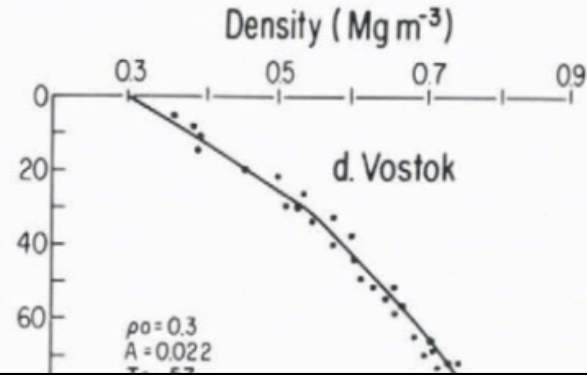
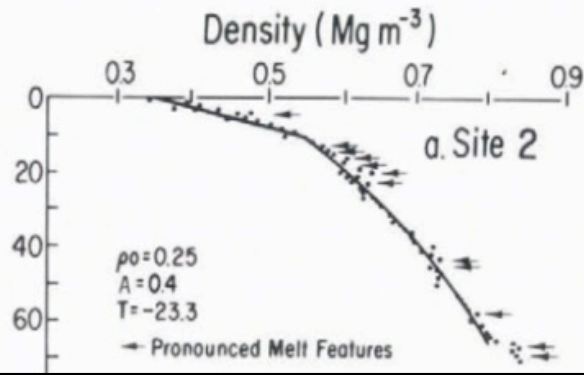
University of Copenhagen, Center for Ice and Climate

An empirical model



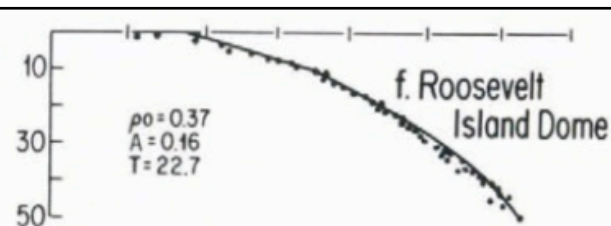
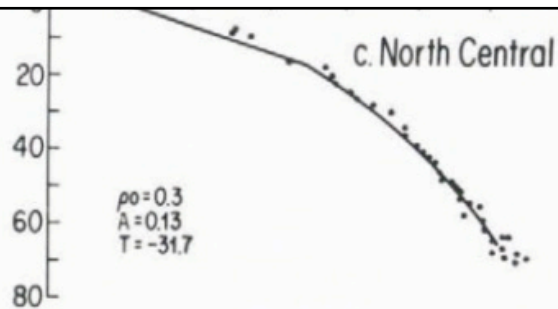
Herron and Langway (1980)

An empirical model



Seems to work, but very little physics in there and it needs tuning each time!

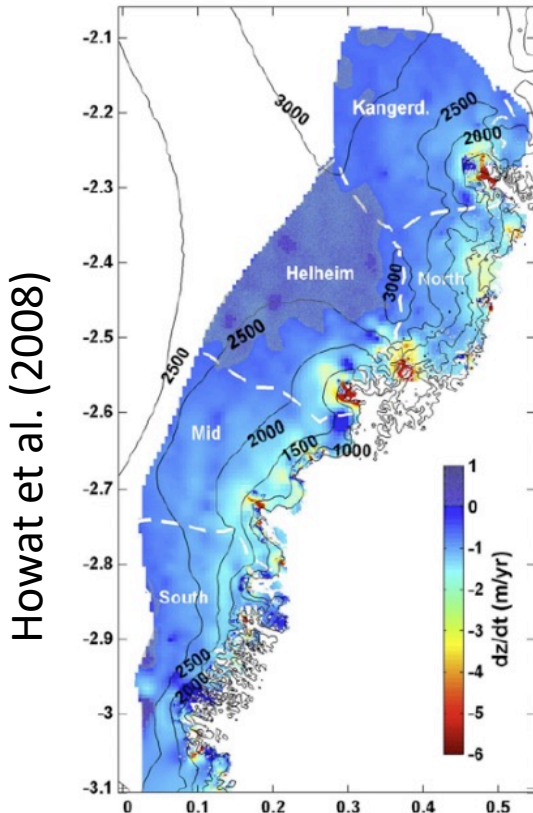
Not very useful for prediction.



Herron and Langway (1980)

Why is getting this right important?

1. Geodetic mass balance



2. Ice core paleo-climate records

Ice cores preserve old ice and old air.

The ice is older than the air by an amount that depends on firn densification rates.



Summary:

- Specific mass balance is the balance between accumulation (mostly snow fall) and ablation (mostly melting) at one point on a glacier.
- We characterize glacier surfaces in terms of several glacier 'facies', related to how much melting happens.
- Firn densification is complex, but important.

On Thursday:

- Our first practical session
- Watch the python videos I sent
- Be prepared to be writing code in class time

References

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